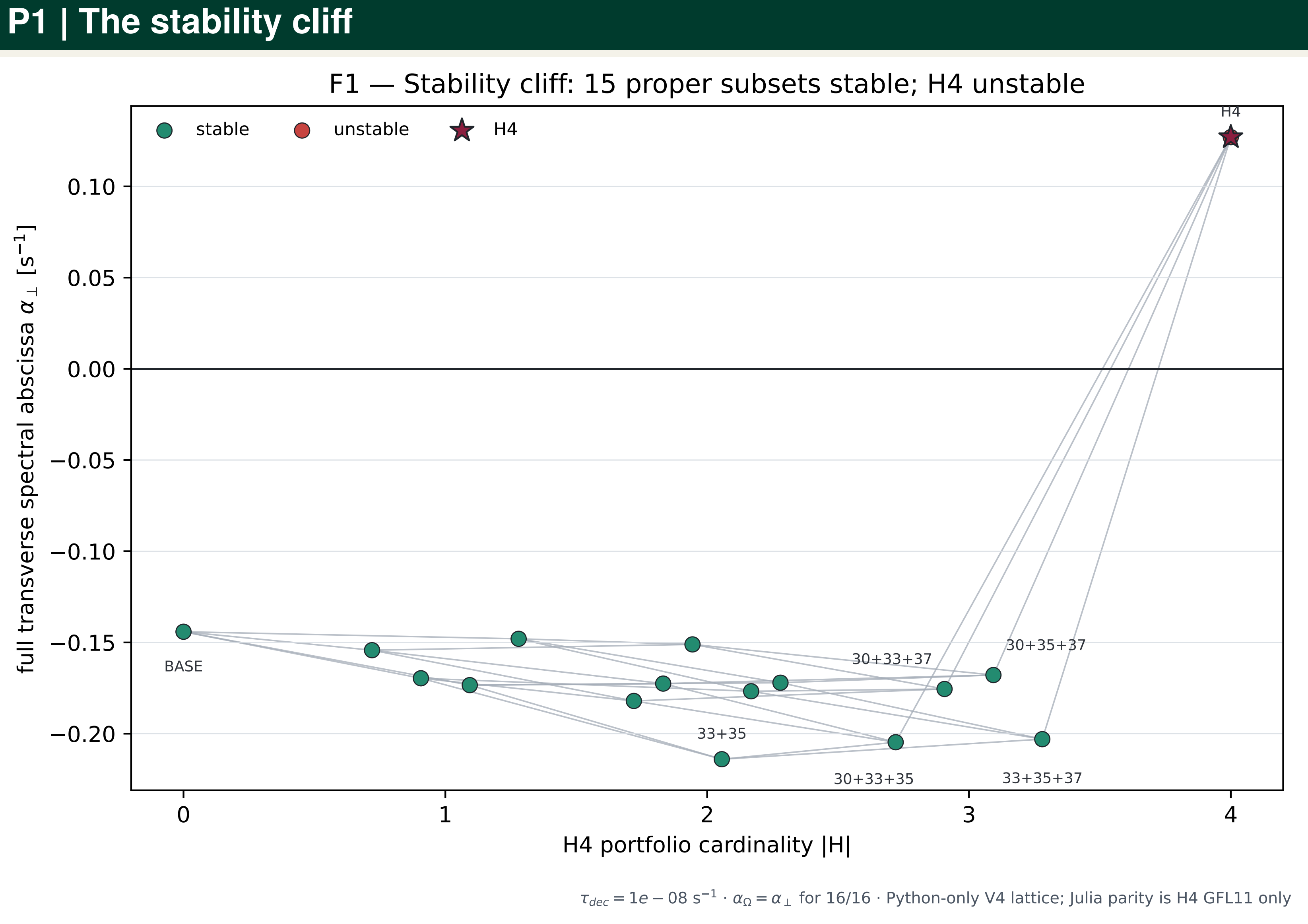


BEYOND NODAL DAMPING

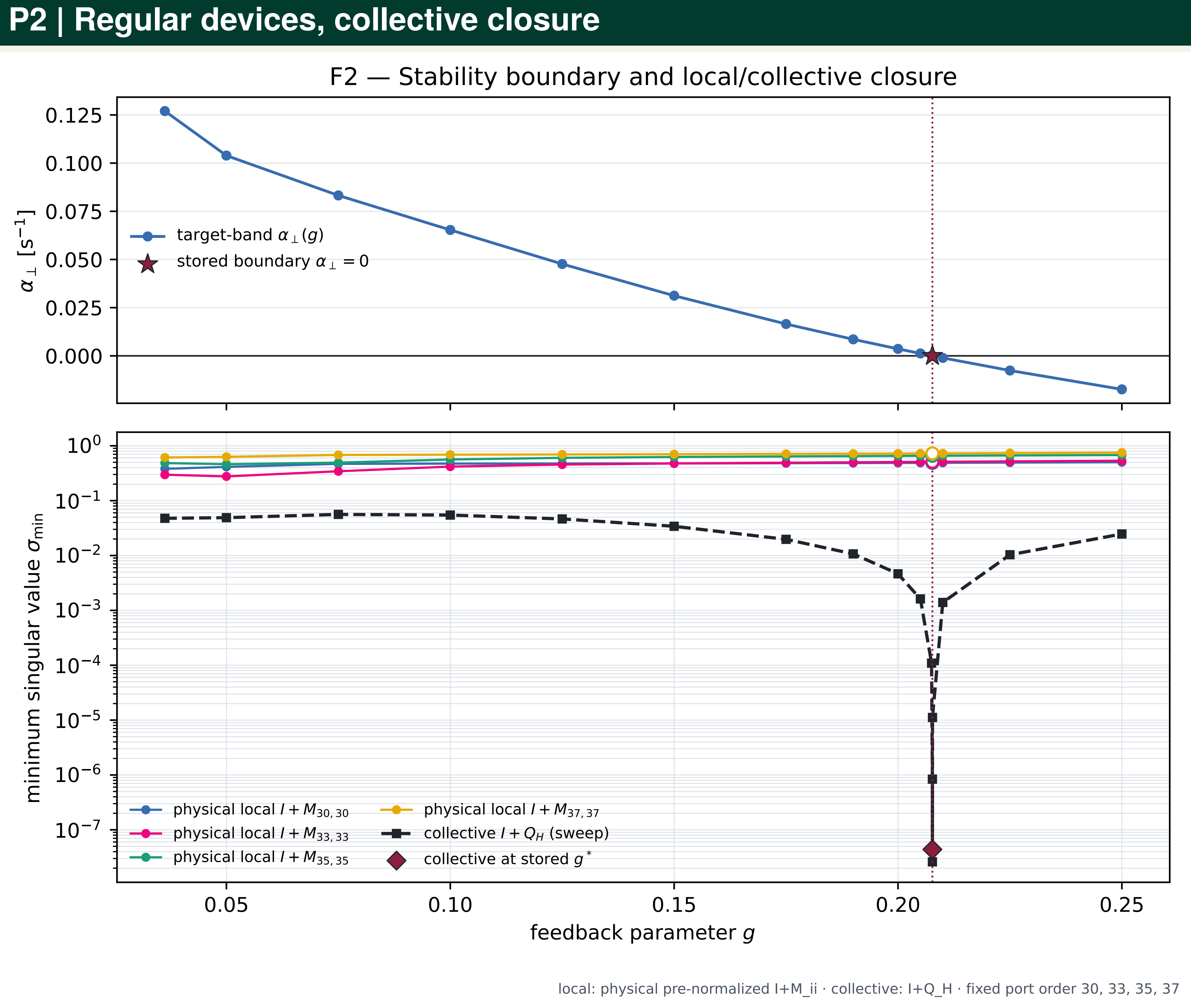
WHEN STABLE REPLACEMENTS FAIL TOGETHER

A canonical IEEE-39 phasor-DAE case, its collective closure, and its operating-point limits

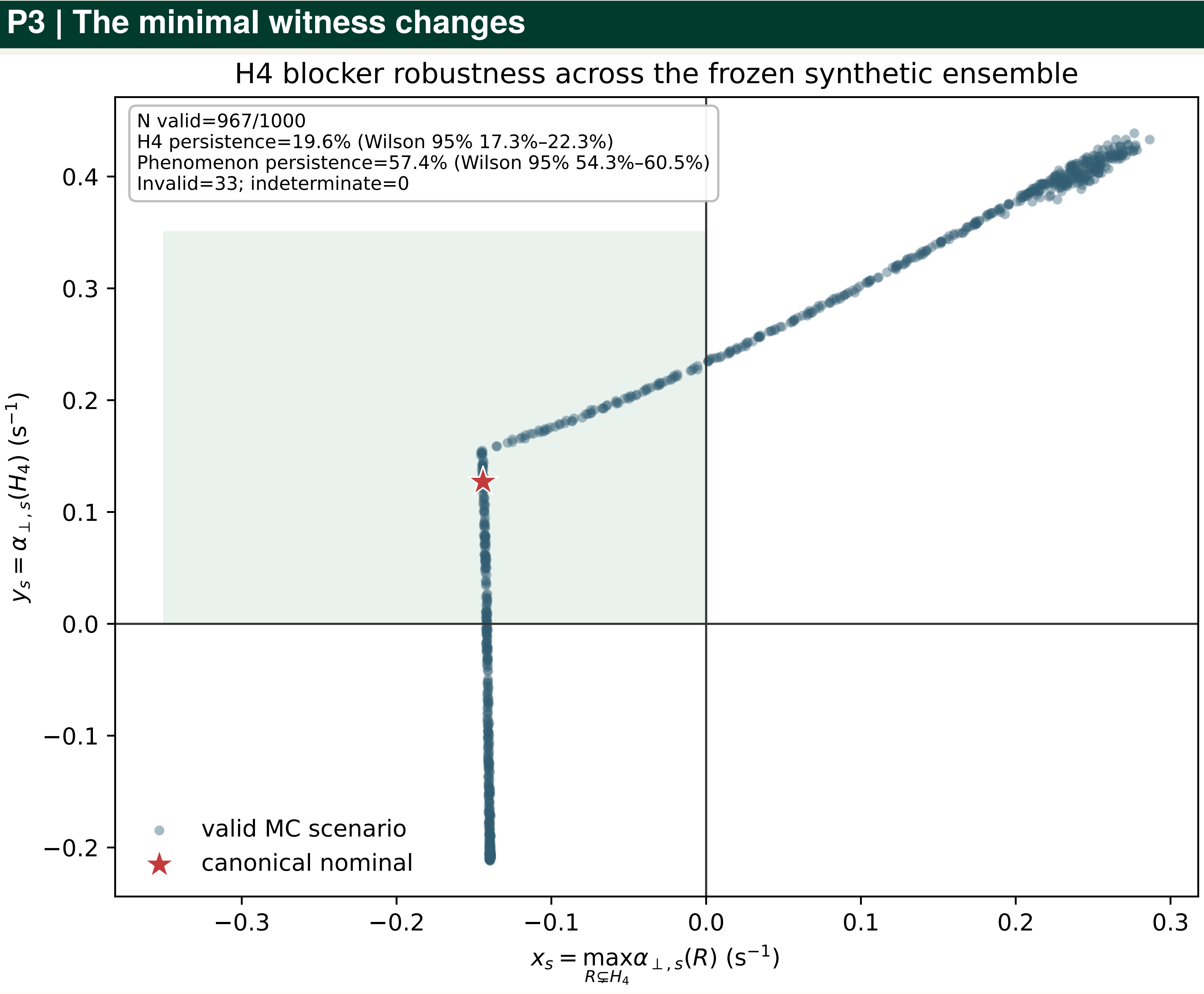
Jorge Luis Mayorga Taborda Universidad de los Andes, Colombia IEEE IAS Annual Meeting 2026 - Vancouver



IEEE-39/TX4 canonical case, deterministic calculation: $H4 = \{30, 33, 35, 37\}$ has 15 stable proper subsets and is unstable ($\alpha_{\perp} = +0.1270\text{ s}^{-1}$, $f = 0.6223\text{ Hz}$). Full transverse spectrum; all 16 exact portfolios; $\tau_{\text{dec}} = 10^{-8}\text{ s}^{-1}$; no ensemble interval. A nominal witness, not a universal blocker.



IEEE-39/TX4 deterministic boundary audit: four physical pre-normalized local factors $I + M_H$ remain nonsingular (minimum $\sigma_{\min} = 0.4892$), while $I + Q_H$ approaches singularity (4.39×10^{-8}) at $g^* = 0.207681$, $\alpha_{\perp} = 0$, and $f = 0.7064\text{ Hz}$. Same equilibrium and port order; no ensemble interval. Local nonsingularity is not positive local damping; strict M1 reduction remains blocked.



V4 blocker atlas: 412 none 190 H4-minimal 365 alternative-witness scenarios
In those 365 cases H4 remains unstable but is no longer minimal. This exact-mask census is descriptive and post hoc; all counts are within $V4 = \{30, 33, 35, 37\}$. Across valid IDs, the three observed minimal triples all include bus 33.

Frozen synthetic envelope: 1,000 scenario IDs; 967 valid and 33 physically infeasible, retained without replacement. H4 minimality: 190/967 (Wilson 95%: 17.3–22.3%); any V4 collective blocker: 555/967 (54.3–60.5%). These are not real-world probabilities.

Reproducibility strip | same-model cross-code reproduction, not independent validation

CROSS-CODE REPRODUCIBILITY

Python ↔ Julia exact GFL11 | 100/100 verdicts agree | max $|\Delta\alpha| = 3.75\text{e-}09\text{ s}^{-1}$ | max $|\Delta f| = 1.53\text{e-}10\text{ Hz}$ | mode identity unresolved

Takeaway: the nominal blocker is not a universal bus set, and the fixed retuning is not a universal cure.
In the audited synthetic V4 envelope, minimal witnesses switch from H4 to triples; at the nominal boundary, physical local factors stay nonsingular while collective closure approaches singularity.

Scope and limits. IEEE-39/TX4 phasor-domain DAE; results depend on model, operating point, and controller policy. IAS26-010 M1 strict is BLOCKED ($2.34 \times 10^{-11} > 10^{-12}$); no η /F3 or GFL/GFM/F4 result is presented. TDS is phasor-domain only, not EMT or field validation. Detailed claim-to-row provenance and the post hoc V4 atlas are in the IAS26-110 supplement.